API Mlab Technical Manual

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1. Introduction

1.1 Overview

API Mlab is a group of API endpoints designed to handle network measurement data, specifically focusing on download data, upload data, and computing aggregated network metrics. The APIs are deployed on Google Cloud Run and leverage Google BigQuery for data storage and querying.

1.2 Components

- **API-Downloads**: Retrieves download network data based on various filters.
- **API-Uploads**: Retrieves upload network data with similar filtering capabilities.
- **API-Compute**: Aggregates and computes combined metrics from both upload and download data.

1.3 Use Cases

- **Network Analysis**: Understanding network performance across different regions and timeframes.
- **Data Integration**: Integrating network metrics into dashboards or other analytical tools.
- **Research and Reporting**: Facilitating research studies and generating reports based on network data.

1.4 Base Urls and Deployment

All APIs are deployed on Google Cloud Run and can be accessed via their respective endpoints:

- **API-Downloads**: `https://mlab-13prsouz.uc.gateway.dev/downloads/`
- **API-Uploads**: `https://mlab-13prsouz.uc.gateway.dev/uploads/`
- **API-Compute**: `https://mlab-13prsouz.uc.gateway.dev/compute/`

2. API Endpoints

2.1 Api-downloads

2.1.1 Overview

The **API-Downloads** endpoint retrieves download network data based on various filters such as country, year, month, region, city, latitude, longitude, AS number, and AS name.

2.1.2 Endpoint Details

- **URL**: `/downloads`
- **Method**: `GET`
- **Description**: Fetches download network data with specified filters and pagination.

2.1.3 Parameters

Parameter	Type	Location	Required	Description
`country`	String	Query	Yes	Filter by country code.
`year`	Integer	Query	No	Filter by year (e.g., 2020).
`month`	Integer	Query	No	Filter by month (1-12).
`region`	String	Query	No	Filter by region within the country.
`city`	String	Query	No	Filter by city where the test was conducted.
`latitude`	Number	Query	No	Filter by latitude of the test location.
`longitude`	Number	Query	No	Filter by longitude of the test location.
`as_number`	Integer	Query	No	Filter by Autonomous System number (ASN).
`as_name`	String	Query	No	Filter by Autonomous System name.
`metric`	String	Query	No	Metric to fetch. Options: `avg_download_speed_mbps`, `avg_packet_loss`, `avg_latency_ms`.
`page`	Integer	Query	No	Pagination: page number. Defaults to `1`.
`page_size`	Integer	Query	No	Pagination: number of results per page. Defaults to `100`. Maximum `5000`.

2.1.4 Responses

- **200 OK**: Successful response with data.
- **400 Bad Request**: Invalid input or query parameters.
- **404 Not Found**: No data found for the given parameters.
- **500 Internal Server Error**: Unexpected server error.

2.1.5 Sample Request

```
GET https://mlab-13prsouz.uc.gateway.dev/downloads/?
country=ZA&year=2023&page=1&page_size=100
```

2.1.6 Sample Response


```
{
  "data": [
    {
      "as_name": "Vodacom",
      "as_number": 29975,
      "avg_download_speed_mbps": 18.6102854803414,
```

```
        "avg_latency_ms": 10.5296666666667,
        "avg_packet_loss": 0.0144136096965,
        "city": "Gauteng",
        "country": "ZA",
        "latitude": -26.1271,
        "longitude": 28.2044,
        "month": 3,
        "region": "Gauteng",
        "year": 2023
      },
      // ... more records
    ],
    "pagination": {
      "has_more": true,
      "page": 1,
      "page_size": 100,
      "total_pages": 607,
      "total_rows": 60637
    }
  }
```


2.2 Api-uploads

2.2.1 Overview

The **API-Uploads** endpoint retrieves upload network data based on various filters similar to **API-Downloads**.

2.2.2 Endpoint Details

- **URL**: `/uploads`
- **Method**: `GET`
- **Description**: Fetches upload network data with specified filters and pagination.

2.2.3 Parameters

Parameter	Type	Location	Required	Description
`country`	String	Query	Yes	Filter by country code.
`year`	Integer	Query	No	Filter by year (e.g., 2020).
`month`	Integer	Query	No	Filter by month (1-12).
`region`	String	Query	No	Filter by region within the country.
`city`	String	Query	No	Filter by city where the test was conducted.
`latitude`	Number	Query	No	Filter by latitude of the test location.
`longitude`	Number	Query	No	Filter by longitude of the test location.
`as_number`	Integer	Query	No	Filter by Autonomous System number (ASN).
`as_name`	String	Query	No	Filter by Autonomous System name.

Parameter	Type	Location	Required	Description
`metric`	String	Query	No	Metric to fetch. Options: `avg_upload_speed_mbps`, `avg_latency_ms`.
`page`	Integer	Query	No	Pagination: page number. Defaults to `1`.
`page_size`	Integer	Query	No	Pagination: number of results per page. Defaults to `100`. Maximum `5000`.

2.2.4 Responses

- **200 OK**: Successful response with data.
- **400 Bad Request**: Invalid input or query parameters.
- **404 Not Found**: No data found for the given parameters.
- **500 Internal Server Error**: Unexpected server error.

2.2.5 Sample Request

```
GET https://mlab-13prsouz.uc.gateway.dev/uploads/?
country=ZA&year=2023&page=1&page_size=100
```

2.2.6 Sample Response


```
{
  "data": [
    {
      "as_name": "Echotel Pty Ltd",
      "as_number": 327693,
      "avg_latency_ms": 27.226125,
      "avg_upload_speed_mbps": 43.8897254794599,
      "city": "Durban",
      "country": "ZA",
      "latitude": -29.8635,
      "longitude": 31.0269,
      "month": 9,
      "num_tests": 32,
      "region": "KwaZulu-Natal",
      "year": 2023
    },
    // ... more records
  ],
  "pagination": {
    "has_more": true,
    "page": 1,
    "page_size": 100,
    "total_pages": 598,
    "total_rows": 59752
  }
}
```

2.3 Api-compute

2.3.1 Overview

The **API-Compute** endpoint aggregates and computes combined metrics from both upload and download data based on specified group-by fields and filters.

2.3.2 Endpoint Details

- **URL**: `/compute`
- **Method**: `GET`
- **Description**: Fetches aggregated network data by combining upload and download metrics based on grouping criteria.

2.3.3 Parameters

Parameter	Type	Location	Required	Description
`group_by`	String	Query	Yes	Comma-separated list of fields to group by (e.g., `year,month,country`).
`country`	String	Query	No	Filter by country code.
`year`	Integer	Query	No	Filter by year (e.g., 2020).
`month`	Integer	Query	No	Filter by month (1-12).
`region`	String	Query	No	Filter by region within the country.
`city`	String	Query	No	Filter by city where the test was conducted.
`latitude`	Number	Query	No	Filter by latitude of the test location.
`longitude`	Number	Query	No	Filter by longitude of the test location.
`as_number`	Integer	Query	No	Filter by Autonomous System number (ASN).
`as_name`	String	Query	No	Filter by Autonomous System name.
`page`	Integer	Query	No	Pagination: page number. Defaults to `1`.
`page_size`	Integer	Query	No	Pagination: number of results per page. Defaults to `100`. Maximum `5000`.

2.3.4 Responses

- **200 OK**: Successful response with aggregated data.
- **400 Bad Request**: Invalid input or query parameters.
- **404 Not Found**: No data found for the given parameters.
- **500 Internal Server Error**: Unexpected server error.

2.3.5 Sample Request

```
GET https://mlab-13prsouz.uc.gateway.dev/compute/?
```

```
group_by=year,month,country&country=ZA&year=2023&page=1&page_size=100
```

2.3.6 Sample Response


```
{
  "data": [
    {
      "avg_download_latency_ms": 58.4369215192295,
      "avg_download_speed_mbps": 39.9355182994538,
      "avg_packet_loss": 0.0582184093455123,
      "avg_upload_latency_ms": 68.8993795674042,
      "avg_upload_speed_mbps": 27.1185748142211,
      "country": "ZA",
      "download_num_tests": 1658340,
      "month": 1,
      "upload_num_tests": 1379683,
      "year": 2023
    },
    // ... more aggregated records
  ],
  "pagination": {
    "has_more": false,
    "page": 1,
    "page_size": 100,
    "total_pages": 1,
    "total_rows": 12
  }
}
```


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3. Data Models

3.1 Download Table Schema

The `download` table in the `MeasurementLab` BigQuery dataset stores download network measurement data.

Field Name	Type	Mode	Description
`year`	INTEGER	NULLABLE	Year of the data (e.g., 2020).
`month`	INTEGER	NULLABLE	Month of the data (e.g., 01 for January).
`country`	STRING	NULLABLE	Country code where the test was conducted.
`region`	STRING	NULLABLE	Region within the country.
`city`	STRING	NULLABLE	City where the test was conducted.
`latitude`	FLOAT	NULLABLE	Latitude of the test location.
`longitude`	FLOAT	NULLABLE	Longitude of the test location.
`as_number`	INTEGER	NULLABLE	Autonomous System number (ASN).
`as_name`	STRING	NULLABLE	Autonomous System name.

Field Name	Type	Mode	Description
`avg_download_speed_mbps`	FLOAT	NULLABLE	Average download speed for the month (Mbps).
`avg_latency_ms`	FLOAT	NULLABLE	Average latency for the month (milliseconds).
`avg_packet_loss`	FLOAT	NULLABLE	Average packet loss rate for the month.
`num_tests`	INTEGER	NULLABLE	Number of tests conducted in that month.

3.2 Upload Table Schema

The `upload` table in the `MeasurementLab` BigQuery dataset stores upload network measurement data.

Field Name	Type	Mode	Description
`year`	INTEGER	NULLABLE	Year of the data (e.g., 2020).
`month`	INTEGER	NULLABLE	Month of the data (e.g., 01 for January).
`country`	STRING	NULLABLE	Country code where the test was conducted.
`region`	STRING	NULLABLE	Region within the country.
`city`	STRING	NULLABLE	City where the test was conducted.
`latitude`	FLOAT	NULLABLE	Latitude of the test location.
`longitude`	FLOAT	NULLABLE	Longitude of the test location.
`as_number`	INTEGER	NULLABLE	Autonomous System number (ASN).
`as_name`	STRING	NULLABLE	Autonomous System name.
`avg_upload_speed_mbps`	FLOAT	NULLABLE	Average upload speed for the month (Mbps).
`avg_latency_ms`	FLOAT	NULLABLE	Average latency for the month (milliseconds).
`num_tests`	INTEGER	NULLABLE	Number of tests conducted in that month.

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4. Error Handling

Proper error handling is crucial for debugging and providing clear feedback to API consumers. Below are the common error responses your APIs may return.

4.1 Error Response Structure

```
{
  "error": "Error message detailing what went wrong."
}
```

4.2 Common Error Codes

Status Code	Description
`400`	Invalid input or query parameters.

Status Code	Description
`404`	No data found for the given parameters.
`500`	Internal server error.

4.3 Example Error Responses

4.3.1 400 Bad Request

- **Cause**: Missing required parameters or invalid parameter formats.
- **Response**:

```json
{
  "error": "Invalid year format. Expected an integer."
}
```

4.3.2 404 Not Found

- **Cause**: No data matches the provided filters.
- **Response**:

```json
{
  "error": "No data found for the given parameters."
}
```

4.3.3 500 Internal Server Error

- **Cause**: Unexpected server-side error.
- **Response**:

```json
{
  "error": "An unexpected error occurred."
}
```

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5. Examples

5.1 Sample Requests

5.1.1 Fetching Download Data for Za in 2023

```
GET https://mlab-13prsouz.uc.gateway.dev/downloads/?
```

```
country=ZA&year=2023&page=1&page_size=50
```

5.1.2 Fetching Upload Data for a Specific as Number

```
GET https://mlab-13prsouz.uc.gateway.dev/uploads/?
country=ZA&as_number=37130&page=2&page_size=100
```

5.1.3 Computing Aggregated Data Grouped by Year and Country

```
GET https://mlab-13prsouz.uc.gateway.dev/compute/?
group_by=year,country&year=2023&page=1&page_size=25
```

5.2 Code Snippets

5.2.1 Python Example Using `requests` Library

```python
import requests

# Define the base URL
base_url = "https://mlab-13prsouz.uc.gateway.dev/downloads/"

# Define query parameters
params = {
    "country": "ZA",
    "year": 2023,
    "page": 1,
    "page_size": 50
}

# Make the GET request
response = requests.get(base_url, params=params)

# Check the response status
if response.status_code == 200:
    data = response.json()
    print("Download Data:", data['data'])
    print("Pagination Info:", data['pagination'])
else:
    print("Error:", response.json()['error'])
```

5.2.2 Curl Example

```
curl -X GET "https://mlab-13prsouz.uc.gateway.dev/downloads/?
country=ZA&year=2023&page=1&page_size=50" -H "Accept: application/json"
```

5.2.3 JavaScript Example Using `fetch`

```
const fetchDownloadData = async () => {
  const url = new URL('https://mlab-13prsouz.uc.gateway.dev/downloads/?
country=ZA&year=2023&page=1&page_size=50');
  const params = { country: 'ZA', year: 2023, page: 1, page_size: 50 };
  url.search = new URLSearchParams(params).toString();

  try {
    const response = await fetch(url, {
      method: 'GET',
      headers: {
        'Accept': 'application/json'
      }
    });

    if (response.ok) {
      const data = await response.json();
      console.log('Download Data:', data.data);
      console.log('Pagination Info:', data.pagination);
    } else {
      const error = await response.json();
      console.error('Error:', error.error);
    }
  } catch (err) {
    console.error('Fetch Error:', err);
  }
};

fetchDownloadData();
```

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6. Glossary

Term	Definition
API	Application Programming Interface, a set of rules and protocols for building software.
BigQuery	Google's fully-managed, serverless data warehouse for large-scale data analytics.
AS Number (ASN)	Autonomous System Number, a unique identifier allocated to each autonomous system on the Internet.
CORS	Cross-Origin Resource Sharing, a mechanism to allow or restrict requested resources on a web server depending on where the HTTP request was initiated.
Pagination	The process of dividing a large dataset into discrete pages to improve performance and usability.

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7. Additional Resources

- **OpenAPI Specification Documentation**: OpenAPI Specification
- **Google Cloud Run Documentation**: Cloud Run Docs
- **Google BigQuery Documentation**: BigQuery Docs
- **MLab Documentation**: Mlab Docs
